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INFRASTRUCTURE REFERENCE · 2026

PROFESSIONAL IMPLEMENTATION BLUEPRINT

Cloudflare Domain, Workers Hosting, DNS & Email

A complete theoretical walkthrough: domain registration, authoritative DNS, static website deployment, staging and production, SSL/TLS, redirects, email forwarding, authentication, operations and recovery.

EDITION	REVIEWED	STATUS
Version 1.1	10 November 2026	Theoretical reference

How to use this guide
Everything uses the reserved illustrative domain `example.com`. Replace example names with your real domain and Worker names. Steps are not live validation; generated DNS values must be copied from your own Cloudflare account. No account or system changes were made in creating this guide.

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1. Executive overview

1.1 Objective

A domain registration is the right to use a name; it is not web hosting or a mailbox. This guide builds a static website and inbound email service without buying a shared-hosting package or a VPS. Cloudflare supplies authoritative DNS, a global HTTPS edge, Workers Static Assets, redirect rules, and inbound Email Routing. A registrar still collects registration/renewal fees; optional sending and mailbox services can cost extra.

Resulting reference configuration

Need	Illustrative value	Responsible service
Purchased domain	`example.com`	Cloudflare Registrar or another registrar
Canonical production website	`https://www.example.com`	Production Worker Custom Domain
Root domain	`https://example.com` redirects to `www`	Proxied placeholder A record + Redirect Rule
Production deployment	`example-production`	Workers Static Assets
Staging deployment	`example-staging`	Separate Worker
Optional private staging URL	`https://staging.example.com`	Staging Custom Domain + Cloudflare Access

Need	Illustrative value	Responsible service
Incoming email	`support@`, `contact@`, `info@`, `admin@` + `example.com`	Cloudflare Email Routing
Email destination	Existing verified Gmail/Outlook/other inbox	Destination mailbox provider
Outgoing email	Optional, not included in forwarding	Cloudflare Email Sending or external provider

1.2 What this does not provide

Cloudflare Workers Static Assets is not PHP shared hosting, a traditional cPanel account, a Windows server, or a general-purpose virtual machine. Uploading a WordPress/PHP application does not make it run as a static site. Cloudflare Email Routing forwards inbound mail; it does **not** create inbox storage, a webmail account, or a general-purpose SMTP sender. If the site requires a database, authenticated back end, server-side rendering, or online store functionality, design those additional services separately.

1.3 Operating principles

1. Keep website DNS in the same Cloudflare account as the Workers that own Custom Domains.
2. Never guess DNS verification tokens, MX priorities, public keys, or assigned nameservers.
3. Prefer distinct staging and production Workers; promote known, tested build output.
4. Let Cloudflare own records created for Worker Custom Domains and Email Routing.
5. Treat the registrar, DNS, web deployment, and email routing as separate failure domains.
6. Make changes incrementally and retain a record of the pre-change state.

2. Architecture and request flow

2.1 Overall architecture

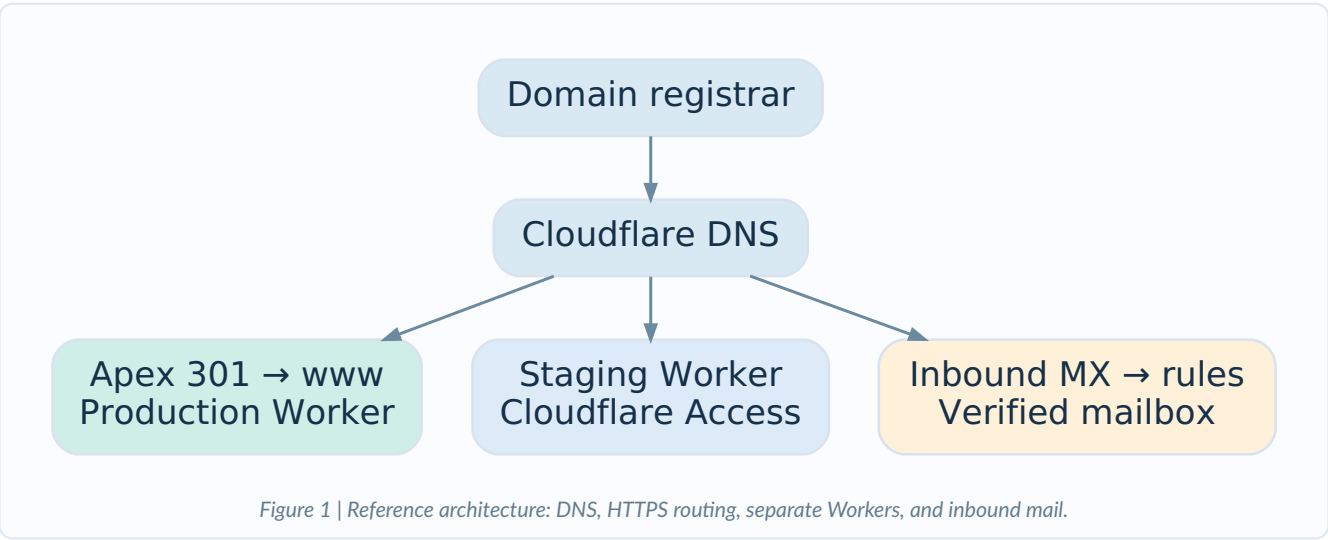


Figure 1 | Reference architecture: DNS, HTTPS routing, separate Workers, and inbound mail.

The diagram is conceptual: DNS looks up hostnames, HTTP traffic reaches Cloudflare's edge, and SMTP traffic reaches Cloudflare's mail exchangers. DNS itself does not forward email messages or serve web pages.

2.2 The configuration workflow



Figure 2 | Recommended provisioning order and acceptance gates.

Go/no-go gates: do not attach production before staging passes; do not change existing inbound MX records before deciding on an email migration; do not enable HSTS or strict DMARC enforcement before complete verification.

3. Prerequisites and decisions

3.1 What you need

- A Cloudflare account with a verified login email and MFA/2FA enabled.
- The ability to register a domain and pay its registration/renewal fees.
- Access to the registrar's nameserver and DNSSEC/DS settings if the domain was bought elsewhere.
- A website folder or the build output of a front-end framework, including `index.html`.
- An existing mailbox that can receive Cloudflare's destination-verification message.
- Optional: a workstation with supported Node.js and Wrangler for reproducible CLI deployments.

3.2 Choose a domain path

If...	Recommended action
You own no domain	Register through Cloudflare Registrar (if available) or another registrar.
You bought the domain at Cloudflare	Cloudflare nameservers are already used; inspect the zone and proceed.
You bought it elsewhere and do not use existing services	Add it to Cloudflare and change authoritative nameservers at the registrar.
You bought it elsewhere and already use email or a website	Inventory records and plan cutover **before** changing nameservers or MX records.

3.3 Choose an email path before onboarding

Requirement	Choose	Important constraint
Receive `support@...` and forward to Gmail/Outlook	**Cloudflare Email Routing**	Inbound only; no independent mailbox or general sending.
Send transactional email from an app	**Cloudflare Email Sending** or another sender	Separate sending service and authentication; prices and eligibility apply.

Requirement	Choose	Important constraint
Have a real `support@...` inbox, folders, webmail and replies	**Hosted mailbox provider**	Use that provider's incoming MX records, not Cloudflare Email Routing MX at the same domain.

4. Register a domain: two routes

4.1 Option A — Buy through Cloudflare Registrar

1. Visit Cloudflare Dashboard and create/sign in to your account.
2. Open **Domain Registration → Register Domains** (the interface may also say **Register domains**).
3. Search the desired name and top-level domain (for example `.com`, `.org`, `.net`).
4. Check availability, registration and renewal prices, restrictions, and supported extensions.
5. Select **Purchase**, choose the registration term, and supply accurate registrant contact information.
6. Review terms, checkout and payment; complete any required email/registrant verification.
7. Open **Domain Registration → Manage Domains** and confirm ownership, expiration date, auto-renewal, and lock/security status.
8. Open the new domain's DNS zone. Cloudflare Registrar domains automatically use Cloudflare authoritative nameservers; **do not change them at a separate registrar**.

Before purchase: Cloudflare Registrar does not support every extension or internationalized domain spelling. Check both availability and transfer/renewal rules before committing. Cloudflare Registrar requires Cloudflare authoritative DNS. [CF-01]

4.2 Option B — Buy through another registrar

1. Register the domain with your preferred registrar, pay and verify contact information.
2. In Cloudflare Dashboard, select **Domains → Onboard a domain / Add a domain**.
3. Enter the **apex name** `example.com`, not `www.example.com`; select the available plan (Free is sufficient for this reference configuration).
4. Review Cloudflare's detected DNS records. Its scan may be incomplete: manually compare existing A, AAAA, CNAME, MX, TXT, SRV, CAA, and relevant subdomain records with the current DNS provider.
5. Cloudflare shows **two assigned nameservers**. Copy the exact values; they look like `<assigned>.ns.cloudflare.com` but are different for each zone.
6. Before delegation, if an old DNSSEC DS record exists at the registrar, perform the documented DNSSEC migration and remove stale DS records at the registrar to prevent validation failures.
7. At your registrar, locate **Nameservers / Custom nameservers**; replace all previous authoritative nameservers with exactly the two assigned by Cloudflare.
8. Save and return to Cloudflare. The zone may temporarily show **Pending Nameserver Update**. Confirm it becomes **Active** before proceeding.
9. Compare all needed records again in the Active Cloudflare zone. Switching nameservers changes **who answers DNS**, not ownership of the domain registration.

Do not: insert Cloudflare's assigned nameservers as ordinary NS records inside the root zone's DNS Records page. For a normal full setup, delegation is made **at the registrar**. [CF-02]

4.3 Nameserver migration checklist

- ☐ Domain registration is valid and contact email verified.
- ☐ Existing website/mail records are exported or copied.
- ☐ Old DNSSEC DS record is handled before the cutover.
- ☐ Exactly the assigned Cloudflare nameservers are registered.
- ☐ Cloudflare zone becomes Active.
- ☐ Existing mail service still resolves and receives mail, if one was present.

5. Activate Cloudflare authoritative DNS

5.1 Identify which records are needed

Open **Cloudflare Dashboard → Domains → example.com → DNS → Records**. The reference design needs only one manually created website A record (for the root redirect) plus a manually managed DMARC TXT record. Cloudflare manages the production/staging Worker Custom Domain records, and Email Routing adds its MX/SPF/DKIM records.

Record	Function	Necessary here?
`A` / `AAAA`	Maps hostname to an IP address	Only a proxied **apex placeholder A** for this design.
`CNAME`	Alias to a hostname	Not manually required for a Worker Custom Domain.
`MX`	Advertises inbound mail exchangers	Yes if using Email Routing; Cloudflare creates them.
`TXT`	Verification and SPF/DKIM/DMARC	SPF/DKIM managed by Cloudflare; DMARC can be added manually.
`NS`	Nameserver delegation	Registrar delegation; additional NS only for special subdomain delegation.
`CAA`	Limits certificate authorities	Optional; incorrect values may block certificates.
`SRV`	Service discovery	Not needed for this static site.
`DS`	DNSSEC chain of trust	Published at registrar/parent; not an ordinary website A record.

5.2 Proxy status rules

- **Orange cloud / Proxied:** Use for Cloudflare-served HTTP/HTTPS traffic and for the apex placeholder that triggers the Redirect Rule.
- **Gray cloud / DNS only:** Typical for external email and verification DNS, when proxy does not apply. MX, TXT, NS and related record types are not Cloudflare HTTP-proxied.
- **Worker-managed:** Do not manually create a competing `www` A or CNAME when attaching the production Worker via **Custom Domain**.

Do not confuse DNS and hosting. A DNS record does not create a web server, an email mailbox, or an SSL certificate by itself.

6. Prepare and deploy the website

6.1 Prepare the site

For a plain static site, prepare files similar to:

```
site/
└── public/
    ├── index.html
    ├── 404.html
    ├── about.html
    ├── contact.html
    ├── css/
    │   └── styles.css
    ├── js/
    │   └── app.js
    └── images/
        └── logo.svg
```

If using React, Vue, Svelte, or another framework, build it and identify the generated `dist`, `build`, or `public` output **appropriate to that framework**. Do not upload secrets, `.env`, `.git`, uncompiled source intended for private use, or credentials. Workers Static Assets hosts the published browser files; a dynamic backend needs a separate design. The legacy **Workers Sites** mechanism is deprecated for new projects; use **Workers Static Assets**. [CF-03]

6.2 Recommended reproducible deployment path (Wrangler)

Wrangler provides exact Worker names and repeatable staging/production releases. On a workstation with supported Node.js and npm, sign in and verify the account:

```
npx wrangler login
npx wrangler whoami
```

Save two illustrative configuration files alongside `public/` in the hypothetical project. These samples describe **assets-only Workers**, so no `main` script is required.

`wrangler.staging.jsonc`

```
{
  "name": "example-staging",
  "compatibility_date": "2026-11-10",
  "assets": { "directory": "./public" }
}
```

`wrangler.production.jsonc`

```
{
  "name": "example-production",
  "compatibility_date": "2026-11-10",
  "assets": { "directory": "./public" }
}
```

Deploy the staging release first:

```
npx wrangler deploy --config wrangler.staging.jsonc
```

After staging acceptance tests, deploy the **same approved artifact** to production:

```
npx wrangler deploy --config wrangler.production.jsonc
```

The commands perform network-authenticated changes to a Cloudflare account. They are **examples only, NOT EXECUTED**. Review the account identity, file paths, intended Worker name, and build output before using them. Current Wrangler accepts assets-only configuration and separate `--config` files. [CF-04]

6.3 Browser-only alternative: Cloudflare Drop

1. Open Cloudflare Drop.
2. Upload the prepared static site folder, ZIP, or accepted website file.
3. Verify the temporary `workers.dev` preview URL and its contents.
4. Complete the **claim** flow within the indicated window (Cloudflare's temporary-account claim documentation specifies 60 minutes).
5. Inspect which Cloudflare **account** now owns the claimed Worker. This may be a claimed temporary-origin account rather than the account containing your registered domain.
6. If the Worker and domain are in different accounts, deploy the static assets again to a Worker in the account that owns the DNS zone. Do **not** assume a claim automatically moves the Worker into an existing DNS-zone account.
7. Create a separate production Worker rather than overwriting the staging preview; verify both resources independently.

Usage recommendation: Drop is convenient for demonstrations and initial previews. For named, versioned, repeatable production deployments, use Wrangler or an authenticated Git/CI deployment. [CF-05]

7. Staging, production, and access protection

7.1 Keep environments separate

Environment	Worker	Public hostname	Change policy
Staging	`example-staging`	`workers.dev` or `staging.example.com`	Deploy first; test and approve.

Environment	Worker	Public hostname	Change policy
Production	`example-production`	`www.example.com`	Only deploy previously approved build.

A `workers.dev` address is generally public by default. To give staging a dedicated hostname, open the staging Worker → **Settings → Domains & Routes → Add → Custom Domain** and enter `staging.example.com`. Cloudflare manages its DNS/edge certificate automatically.

7.2 Restrict staging access

1. Confirm Cloudflare **Zero Trust** is enabled for the account and that you have permissions to manage Workers and Access applications.
2. Open **Workers & Pages → example-staging → Access**.
3. Select **Protect this Worker behind Access**.
4. Choose **All traffic** (not **Previews only**). Previews only does not cover the staging Worker's production URL or its `staging.example.com` Custom Domain.
5. Under **Authentication policy**, permit only your approved testers or identities; review the session duration.
6. Select **Apply Access** and confirm that the policy is enabled for **example-staging** only. The production Worker should remain publicly available.
7. Test an approved identity and an unauthorized/incognito session against **both** the Worker's `workers.dev` hostname and `staging.example.com` (if configured), including any preview URLs.

Worker-level **All traffic** protection covers the Worker's routes, Custom Domains, `workers.dev` hostname, and previews. It does not currently support WebSocket upgrades; if your staging application uses WebSockets, use appropriate hostname-based Access policies and test every reachable entry point instead. [CF-06]

7.3 Staging acceptance gate

Verify navigation, links, forms, static file paths, mobile layout, accessibility, browser console, expected 404 behavior, cache assumptions, and any external APIs. A contact form that sends email is **not** implemented merely by hosting a static HTML form; it requires a form endpoint/Worker and abuse controls or a third-party service.

8. Attach the production custom domain

After the production Worker is deployed **in the account owning the Active DNS zone**:

1. Open **Workers & Pages → example-production**.
2. Open **Settings → Domains & Routes**.
3. Select **Add → Custom Domain**.
4. Enter `www.example.com` and confirm.
5. Wait for Cloudflare to provision the hostname's DNS mapping and edge certificate.
6. Verify the Custom Domain is Active/ready, then open `https://www.example.com`.

A **Worker Custom Domain** makes the Worker the origin for the exact hostname. It is different from a Worker **Route**, which commonly places a Worker in front of an existing origin. Do not attach only `example.com` and expect `www.example.com` to work: the hostnames are distinct. [CF-07]

Never add a manual competing `www` A or CNAME on the assumption the Worker requires a fake origin. Cloudflare creates and maintains the DNS mapping for the Worker Custom Domain.

9. Create the root-domain redirect

9.1 Add a proxied apex placeholder

Go to **Domains → example.com → DNS → Records → Add record** and use:

Field	Value
Type	`A`
Name	`@` (apex/root)
IPv4 address	`192.0.2.0`
Proxy status	**Proxied** (orange cloud)
TTL	Auto

This is an **originless redirect placeholder** reserved for documentation use, not an accessible web server. Because the record is proxied, Cloudflare receives the HTTP request and can apply a redirect. If an existing apex A/AAAA/CNAME already serves something important, inventory it and plan a migration before replacement. [CF-07]

9.2 Create a redirect rule

1. In the domain zone, open **Rules → Overview → Create rule → Redirect Rule** (or the dashboard's current equivalent).
2. Name it `Redirect apex to www`.
3. Under **When incoming requests match**, choose **Custom filter expression**.
4. Enter the condition below.
5. Under **Then**, choose **Dynamic URL redirect**, enter the target expression below, set **301**, and enable **Preserve query string**.
6. Save/deploy the rule and test it with a browser and an HTTP client.

Condition

```
(http.host eq "example.com")
```

Dynamic target expression

```
concat("https://www.example.com", http.request.uri.path)
```

Intended outcomes

Incoming URL	Final canonical URL
`http://example.com/`	`https://www.example.com/`
`https://example.com/about.html`	`https://www.example.com/about.html`
`https://example.com/contact.html?src=qr`	`https://www.example.com/contact.html?src=qr`

The path is preserved by the expression and the query string by the setting. HTTP-to-HTTPS enforcement may cause a separate intermediate redirect. Do not add the reverse redirect `www → apex`, or the two rules may loop. A 301 redirect can change a POST to GET in some clients; do not use canonical host redirection as an API method-preservation mechanism without considering 307/308. [CF-08]

10. Configure HTTPS and DNSSEC

10.1 Edge certificates and HTTPS

1. In **Workers & Pages → production Worker → Domains & Routes**, verify `www.example.com` is active and covered by a certificate.
2. In **example.com → SSL/TLS → Edge Certificates**, confirm the applicable certificates are active for apex and `www`.
3. Find **Always Use HTTPS** and enable it after certificate readiness.
4. Test `http://example.com`, `https://example.com`, `http://www.example.com`, and `https://www.example.com`.
5. All four should end at `https://www.example.com`, allowing intermediate redirects.
6. **Do not enable HSTS until all relevant HTTPS hostnames work.** HSTS can persist in browsers and complicate rollback.

This is an originless Worker deployment. Do not invent an origin server or install a server certificate on an address you do not control. If you later connect an external origin, re-evaluate SSL/TLS mode and origin verification; avoid weak configurations such as Flexible for a real HTTPS origin. [CF-09]

10.2 DNSSEC

For a Cloudflare Registrar domain, use the registrar's **Manage Domains → Configuration → DNSSEC** controls. For a third-party registrar domain, first confirm delegation is stable and the zone is Active, then open Cloudflare **DNS → Settings → DNSSEC**. Enable DNSSEC and publish the Cloudflare-provided DS information at the registrar if the registrar does not automate it. Verify the chain of trust. A stale DS entry during a nameserver migration can cause widespread resolution failure. [CF-10]

11. Set up incoming email forwarding

11.1 Email delivery architecture

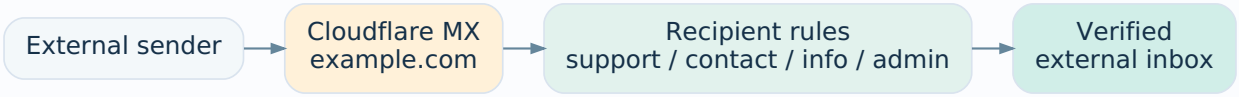


Figure 3 | Email Routing forwards messages to an existing verified inbox.

Important: Creating a routing address is not creating a mailbox. Messages are forwarded to an existing verified destination. Replying **as** `support@example.com` is a separate outbound-service problem.

11.2 Onboard the domain

1. Open Cloudflare Dashboard → **Compute → Email Service → Email Routing**.
2. Select **Onboard Domain** and choose `example.com`.
3. Review the MX records and the SPF/DKIM TXT records Cloudflare proposes.
4. Confirm onboarding. This writes the necessary records to Cloudflare DNS.
5. Open **Email Routing → Settings → DNS records** and verify the records show configured.

Before approving, ensure an existing external email provider does **not** already depend on the apex MX records. Email Routing and a separate full mailbox provider should not be given competing independent apex MX sets. DNS records may take time to propagate globally. [CF-11]

11.3 Verify the destination address

1. Open **Compute → Email Service → Email Routing → Destination Addresses**.
2. Add an actual existing inbox, such as a personal Gmail or Outlook address that you control.
3. Open the Cloudflare verification message delivered there and follow its confirmation link.
4. Ensure the destination status is **Verified** before making forwarding rules.

11.4 Create four routing rules

In **Email Routing → Routing Rules → Create routing rule**, add each **local part** (the part before `@`) and select **Send to an email** with the verified destination.

Rule name / pattern	Full address	Action	Destination
`support`	`support@example.com`	Send to an email	Verified external inbox
`contact`	`contact@example.com`	Send to an email	Verified external inbox
`info`	`info@example.com`	Send to an email	Verified external inbox
`admin`	`admin@example.com`	Send to an email	Verified external inbox

1. Create `support`; save and confirm active.
2. Repeat for `contact`, `info`, and `admin`.

3. Send a separate message to each address **from an unrelated external sender**.
4. Check the destination inbox, spam folder, and Cloudflare Email Routing Activity Log.
5. Leave **Catch-all** off unless you need mail sent to every otherwise-unconfigured local part; catch-all often increases spam volume.

There are **no separate DNS records for each local part**. The same domain-level MX records receive all mail; routing rules dispatch it by recipient address. [CF-12]

12. SPF, DKIM, and DMARC

Email authentication has three complementary pieces. SPF authorizes envelope-sender hosts, DKIM cryptographically signs messages, and DMARC requires aligned authenticated identity and supplies a policy. They do not individually guarantee inbox delivery or prevent all spoofing.

12.1 SPF — managed by Email Routing

Cloudflare's documented root-domain Email Routing SPF string is:

```
v=spf1 include:_spf.mx.cloudflare.net ~all
```

The TXT record's name is `@` / `example.com`. **Do not add it again if onboarding created it.** If additional sending providers need SPF authorization, merge mechanisms into **one** valid SPF TXT record at a hostname; two separate `v=spf1` TXT policies at the same name can produce a permerror. Confirm the service's actual DNS instructions before editing SPF. [CF-13]

12.2 DKIM — managed by Email Routing

Cloudflare Email Routing's documented DKIM selector is:

```
cf2024-1._domainkey.example.com
```

The TXT value contains a **Cloudflare-generated public key**, not a copyable fixed sample. Do not type `p=...` or invent a key. Other providers may use different selectors and can coexist if configured correctly. [CF-13]

12.3 DMARC — add a monitoring policy

For a new domain with no verified sending setup, begin with DMARC monitoring:

DNS field	Value
Type	`TXT`
Name	`_dmarc`
Content	`v=DMARC1; p=none`
TTL	Auto

Optionally, establish a working `dmarc@example.com` destination and then add aggregate reporting:

```
v=DMARC1; p=none; rua=mailto:dmarc@example.com
```

Only after legitimate senders are authenticating and aligning should you consider `p=quarantine`, then `p=reject`. Avoid multiple conflicting TXT DMARC records at `_dmarc`. Reports are frequently XML and may require a processing service. [CF-14]

13. Outbound email and full mailbox alternatives

13.1 Option A — Cloudflare Email Sending

Cloudflare Email Sending is a separately onboarded service (documented as beta in the current Email Service interface). It is distinct from Email Routing. It can send application/transactional messages via Workers, API, or authenticated SMTP. The documented SMTP endpoint is `smtp.mx.cloudflare.net` on port `465` with implicit TLS, username `api_token`, and an API token with the required Email Sending permission. Do not embed that token in public JavaScript or downloadable website files. Sending to arbitrary recipients requires the Workers Paid plan under Cloudflare's documented pricing. Check feature availability and terms before implementation. [CF-15]

To onboard: open **Compute → Email Service → Email Sending → Onboard Domain**, select the domain, approve the sending-specific authentication and bounce-processing DNS records, and verify them in **Email Sending → Settings**. Cloudflare uses separate `cf-bounce` MX/SPF and `cf-bounce._domainkey` DKIM names; these **do not replace** the apex Email Routing MX records. The sending service is **not a webmail inbox**.

13.2 Option B — Full hosted mailboxes

If you need real inboxes and ordinary replies (for example, four separate business mailboxes), subscribe to a mailbox provider such as Google Workspace, Microsoft 365, or Zoho Mail, and follow its domain onboarding process:

1. Add your domain to the mail provider and complete ownership verification TXT records.
2. Provision the mailboxes and users.
3. Plan the cutover, including existing messages, aliases, and backups.
4. Replace the apex incoming MX set with the provider-supplied set; **do not keep Cloudflare Email Routing MX as a competing independent set**.
5. Configure the provider's SPF, DKIM, and DMARC as directed; merge SPF authorization where appropriate.
6. Verify inbound, outbound, spam handling, calendars/clients (if applicable), and mailbox access.

Cloudflare DNS and Workers hosting may remain unchanged while email is hosted elsewhere. Do **not** copy example Google or Microsoft MX records without obtaining the actual values from the provider's administration console.

13.3 Email decision summary

Feature	Email Routing	Email Sending	Hosted mailbox
Receives mail for domain	Forwards incoming	No inbound mailbox	Yes
Stores user inbox	No	No	Yes
Sends to arbitrary recipients	No	Yes, subject to plan/limits	Yes
Typical use	`support@` → existing inbox	App notifications	Business correspondence

Feature	Email Routing	Email Sending	Hosted mailbox
Root-domain inbound MX	Cloudflare	Does not set inbound MX itself	Mailbox provider

14. Complete DNS inventory

The inventory is **scenario-specific**. DNS records are not a shopping list: do not create every optional record automatically. Record names are shown relative to `example.com`; `@` represents the apex. The actual Cloudflare-generated values, MX priorities, DKIM public key, and hostname mappings must be taken from the account's dashboard.

14.1 Baseline: Workers + www canonical + Email Routing

Type	Name	Value / destination	Proxy	Who creates it	Required?
A	`@`	`192.0.2.0`	**Proxied**	You	Yes, for apex redirect
Worker Custom Domain	`www`	`example-production`	Managed	Cloudflare	Yes
Worker Custom Domain	`staging`	`example-staging`	Managed	Cloudflare	Optional
MX	`@`	`route1.mx.cloudflare.net`	N/A	Email Routing	Yes for Routing
MX	`@`	`route2.mx.cloudflare.net`	N/A	Email Routing	Yes for Routing
MX	`@`	`route3.mx.cloudflare.net`	N/A	Email Routing	Yes for Routing
TXT	`@`	`v=spf1 include:_spf.mx.cloudflare.net ~all`	N/A	Email Routing	Yes for Routing
TXT	`cf2024-1._domainkey`	Account-generated DKIM public key	N/A	Email Routing	Yes for Routing
TXT	`_dmarc`	`v=DMARC1; p=none`	N/A	You	Strongly recommended

MX priorities: use the values shown by Cloudflare during onboarding; do not invent them. Each automatically provisioned Worker Custom Domain includes DNS mapping and certificate management, but it may not appear as a plain CNAME in the UI. Never add a conflicting `www` A/CNAME manually. [CF-07] [CF-13]

14.2 Additional records for optional Email Sending

Type	Name	Function	Owner
MX (multiple)	`cf-bounce`	Bounce processing	Email Sending
TXT	`cf-bounce`	Sending SPF	Email Sending
TXT	`cf-bounce._domainkey`	Sending DKIM	Email Sending
TXT	`_dmarc`	Single DMARC policy, if managed/proposed	Coordinate with existing DMARC

Do not copy these record names as proof of a configured sender; activate the product and verify each actual record in Email Sending settings. [CF-13]

14.3 Records for other DNS services (only if needed)

Type	Example use	Caution
`TXT`	Third-party domain verification	Copy exact token from issuing service.
`CNAME`	SaaS service alias / external app	Avoid conflicts with Worker-managed hostname.
`CAA`	Limit certificate issuers	Can prevent Cloudflare certificate issuance.
`SRV`	SIP, gaming or other service discovery	Use provider-specific host/port/priority.
`NS`	Delegate `sub.example.com` elsewhere	Different from apex registrar nameserver change.
`DS`	DNSSEC chain at registrar	Preserve matching key and digest.

14.4 Common conflicting-record mistakes

- Manual `A www → 192.0.2.0` **and** a `www` Worker Custom Domain: incorrect for this design.
- `CNAME www → worker.account.workers.dev` **and** a `www` Worker Custom Domain: unnecessary and can conflict.
- `A mail → 192.0.2.0`: does **not** create a mail server.
- Duplicate SPF TXT policies at `@`: SPF can fail with `permerror`.
- MX from Cloudflare Routing **and** external mailbox provider at the same domain: mail delivery becomes unpredictable.
- Public staging route with no Access policy: staging content may leak.
- A placeholder A record set to **DNS only**: redirect rule cannot intercept it through the Cloudflare proxy.

15. Validation and acceptance tests

Evidence labels: `EXECUTED` = actually performed in the named runtime/account; `STATICALLY CHECKED` = reviewed against documents or code; `SIMULATED` = exercised in a mock/model; `NOT TESTED` = no such test occurred. This theoretical guide is **STATICALLY CHECKED** for the described platform procedures. Live account and network outcomes are **NOT TESTED**.

15.1 DNS commands (illustrative)

On Windows with `nslookup`:

```
nslookup -type=NS example.com
nslookup -type=MX example.com
nslookup -type=TXT example.com
nslookup -type=TXT _dmarc.example.com
nslookup -type=TXT cf2024-1._domainkey.example.com
```

On a system with `dig`:

```
dig NS example.com
dig MX example.com
dig TXT example.com
dig TXT _dmarc.example.com
```

When `@` is a proxied A placeholder, public DNS typically returns **Cloudflare edge IPs**, not `192.0.2.0`; that is expected. A healthy DKIM lookup returns a full key, not just `p=...`.

15.2 HTTP redirect commands (illustrative)

```
curl.exe -I https://www.example.com
curl.exe -I https://example.com/about.html?src=guide
curl.exe -IL http://example.com
```

From Linux/macOS, use `curl` instead of `curl.exe`. Verify status and `Location` headers, follow the chain, and confirm the response body and static files. A path may still return 404 if the asset does not exist; redirects do not create missing pages.

15.3 Acceptance test matrix

ID	Test	Pass criterion
D-01	Registrar delegation	Cloudflare-assigned NS are authoritative; zone Active.
D-02	DNSSEC after activation	DS matches Cloudflare key; DNS resolves without validation failures.
W-01	Staging	Correct staging content and expected asset paths load.

ID	Test	Pass criterion
W-02	Staging privacy	Approved identity works; unauthorized identity is denied.
W-03	Production	`https://www.example.com` serves approved Worker release.
W-04	Root redirect	Apex redirects to `https://www`, preserving valid paths and query strings.
W-05	HTTPS	HTTP ultimately uses HTTPS; relevant certificates active.
W-06	Errors	Nonexistent asset returns expected 404; no sensitive file leaks.
E-01	Email DNS	MX and single valid SPF, DKIM selector, DMARC are present.
E-02	Destination	Existing destination address is verified.
E-03	Four routing rules	`support`, `contact`, `info`, `admin` each deliver externally originated test mail.
E-04	Optional sending	External recipient receives authorized signed email; SPF/DKIM/DMARC assessed.
O-01	Repeated release	Redeploying approved build does not change DNS or create duplicate Workers.
O-02	Recovery	Prior deployed release can be selected/ rolled back within supported limits.

15.4 Negative and boundary tests

Also test incorrect hostname, missing asset, extra query parameters, unusual URL encoding, expired staging authentication, an unverified email destination, a mailbox-provider MX conflict, and a repeated deployment. Keep test emails distinguishable with unique subjects and review the Routing Activity Log and destination spam folder. Do not weaken DNSSEC or DMARC just to suppress a symptom without isolating the root cause.

16. Release workflow and recovery



Figure 4 | Controlled release workflow; rollback is a separate recovery step.

16.1 Routine change procedure

1. Version the source code or archive the source/build inputs.
2. Build once and identify the exact output to be promoted.

3. Deploy to `example-staging`, verify it is the intended account and Worker.
4. Run functional, security, and compatibility tests; obtain approval.
5. Deploy that approved build to `example-production`.
6. Smoke-test `www.example.com`, root redirect, key assets, and any form/API integration.
7. Record deployment ID, date, author, and change summary.
8. Monitor error rates and traffic, and retain the previous known-good release.

16.2 Recovery and rollback

Use **Workers & Pages → Worker → Deployments** to inspect versions and use the supported rollback control to restore a prior Worker release. Rolling back a Worker may **not** revert DNS changes, Email Routing changes, database schema/data, external APIs, or third-party storage. Plan recovery for each component separately. [CF-16]

Change safety: Do not remove records as a blanket reset. Before a DNS/email migration, export or screenshot prior settings, lower TTL only where supported and useful, schedule a rollback window, and ensure the previous destination can receive mail if the cutover must be reversed.

17. Security, limits, and operations

17.1 Minimum hardening

Control	Suggested action
Identity	Enable Cloudflare account 2FA; use least-privilege roles and API tokens.
Domain ownership	Verify registrant email and auto-renewal; monitor expiration.
DNS	Audit existing records; enable DNSSEC after proper delegation.
TLS	Use HTTPS; enable Always Use HTTPS after certificate checks.
Staging	Use Worker-level Access and test all alternate entry paths.
Website	Never publish secrets or private config in static assets.
Headers	Consider CSP, HSTS (only after validation), X-Content-Type-Options, referrer policy.
Email	Use SPF/DKIM/DMARC, verified destinations, and least-privilege sending tokens.
Monitoring	Inspect Worker deployments/logs, DNS health, and Email Routing Activity Log.
Backups	Keep original files, build output, release notes, and DNS change history.

17.2 Usage and costs

- Domain registration and renewal are paid (price depends on registrar/extension).
- Cloudflare's Free tier can be suitable for a basic static website, with platform and account limits.

- Requests to Workers Static Assets are documented as free/unlimited, while Worker **script** execution has separate plan quotas; optional features may be charged. [CF-17]
- Cloudflare Email Routing offers free inbound forwarding; outbound sending to arbitrary recipients depends on a paid plan and service limits. [CF-18]
- Confirm current limits before large traffic, large files, video range requests, transactional email, or dynamic workloads. Static assets are not unlimited storage for every imaginable use case.

17.3 Operational responsibilities

A site without traditional hosting still needs domain renewal, content updates, credential handling, security review, email-delivery testing, and release rollback planning. If you add a database, payments, authentication, dynamic workers, or mail sending, the architecture and threat model expand.

18. Troubleshooting and failure isolation

Symptom	Most likely layer	First checks	Avoid
Zone stays Pending	Registrar / delegation	Assigned NS, extra NS, old DS, propagation	Repeated random DNS edits
All DNS lookups fail with DNSSEC	Registrar DS mismatch	DNSSEC chain and DS record	Deleting arbitrary zone data
`www` does not serve site	Worker custom domain	Account, Worker name, hostname/cert status	Manual competing `www` CNAME
Root reaches error instead of redirect	Proxied placeholder / rule	Apex A Proxied, rule enabled/ match	Using DNS-only placeholder
Redirect loop	URL rules	Rule direction, destination host, HTTPS interactions	Two opposite canonical rules
HTTPS certificate error	SSL/TLS	Certificate issuance, hostname, DNS ownership	Immediate HSTS enforcement
Missing CSS/JS/images	Static assets	Built paths, case sensitivity, output folder	Assuming host creates files
Staging is publicly readable	Access	All staging hostnames, Worker-level policy	Relying on an obscure URL
Inbound email fails	Mail DNS/rules	MX, destination verified, rule status, Activity Log	Mixing incompatible MX providers
Mail unexpectedly spammed	Sender/auth/forwarding	Message headers, SPF/DKIM/ DMARC, destination filter	Assuming DNS alone guarantees inbox
Sending email fails	Outbound provider	Plan, verified sender, token permission, SMTP config	Exposing token client-side
Old site appears	Cache/version	Worker deployment, asset cache, correct hostname	Purging blindly before verifying deployment

Escalation order: (1) registrar/NS → (2) Cloudflare Active zone → (3) DNS records → (4) Worker/redirect or mail service → (5) TLS/authentication → (6) client/browser/inbox. Record exactly which hostname, time, request, sender, and error failed.

19. End-to-end completion checklist

Domain and account

- ☐ Cloudflare account created; verified login and 2FA enabled.
- ☐ Domain registered; expiry, payment, renewal and contact verification reviewed.
- ☐ If externally registered: Cloudflare nameservers applied at registrar.
- ☐ Old DNS records inventoried and old DS handled before cutover.
- ☐ Cloudflare DNS zone Active and delegation verified.

Website, DNS, staging, production

- ☐ Site artifact prepared; no secrets in browser-delivered assets.
- ☐ Separate staging Worker deployed and tested.
- ☐ Staging access restricted if private; no unprotected alternate URL.
- ☐ Separate production Worker in the same account as domain zone.
- ☐ `www.example.com` attached as production Worker Custom Domain.
- ☐ Apex A record `@ → 192.0.2.0` is Proxied.
- ☐ Root-to-www Redirect Rule deployed; query/path tested.
- ☐ HTTPS certificates active; Always Use HTTPS enabled.
- ☐ DNSSEC enabled and verified if applicable.

Email

- ☐ Chosen **Email Routing** versus **hosted mailbox** path (not conflicting apex MX sets).
- ☐ Email Routing onboarding completed where selected.
- ☐ Destination email address verified.
- ☐ Four individual routing rules enabled.
- ☐ MX, SPF, DKIM and DMARC reviewed for duplicates/conflicts.
- ☐ Four externally originated messages received in intended destination.
- ☐ Outbound email separately configured and tested only if required.

Operations

- ☐ HTTP and DNS acceptance tests completed with actual results recorded.
- ☐ Restore/rollback plan recorded; latest known-good release retained.
- ☐ Account, registrar and destination inbox recovery methods documented.
- ☐ Applicable service limits, charges, notifications and renewal dates reviewed.

20. Glossary and official references

20.1 Terms

Apex/root: Bare domain such as `example.com`. **Registrar:** Seller/manager of domain registration; not necessarily the DNS provider. **Authoritative DNS:** Nameservers that publish DNS answers. **Workers Static Assets:** Cloudflare-hosted files served from its edge. **Custom Domain:** A hostname attached directly to a Worker that acts as its origin. **Proxied:** Cloudflare handles eligible HTTP requests before any origin connection. **MX:** Mail exchanger records for inbound SMTP. **Email Routing:** Incoming mail forwarding, not mailbox hosting. **SPF/DKIM/DMARC:** Sender authentication and policy technologies. **DS:** DNSSEC delegation record stored at the parent/registrar side. **Staging:** Pre-production testing environment. **Canonical hostname:** Preferred public URL (`www` in this guide).

20.2 Official documentation (reviewed for this edition)

- **[CF-01]** Cloudflare Registrar: Register a new domain
- **[CF-02]** DNS: Set up a primary zone (full setup)
- **[CF-03]** Workers Static Assets: Get started
- **[CF-04]** Wrangler configuration and Static assets configuration
- **[CF-05]** Claim deployments and temporary accounts
- **[CF-06]** Cloudflare Access for Workers
- **[CF-07]** Worker Custom Domains and placeholder apex records
- **[CF-08]** Single Redirects and settings
- **[CF-09]** Always Use HTTPS
- **[CF-10]** DNSSEC overview
- **[CF-11]** Email Routing onboarding
- **[CF-12]** Email Routing rules and addresses
- **[CF-13]** Email Service domain DNS records and authentication
- **[CF-14]** Email security records and DMARC
- **[CF-15]** Email Sending and SMTP reference
- **[CF-16]** Worker rollback
- **[CF-17]** Static Assets billing and limitations
- **[CF-18]** Email Service pricing

20.3 Revision history

Edition	Date	Change
1.0	10 November 2026	Initial theoretical reference guide.
1.1	10 November 2026	Clarified staging Access coverage and correct selection, normalized Mermaid line breaks, noted Email Sending beta, and updated the Worker rollback reference URL.

Document control: Version 1.1 · 10 November 2026 · Theoretical reference only. Technical platform steps were cross-checked against Cloudflare's public documentation. No account access, deployment, actual DNS, mail routing, or native OS test was performed. Always confirm account-specific generated values and current Cloudflare interface labels before making changes.